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REVIEW ARTICLE

Employing microbotox technique for facial rejuvenation and face-lift

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Abstract

Background: Microbotox technique, intradermal injection of microdroplets of botulinum toxin, is a favorable rejuvenation approach in subjects who prefer more natural appearance.

Aim: To determine the best injection techniques (dosage, concentration, as well as number and location of injection points), efficacy and side effects of this innovative technique for facial rejuvenation.

Methods: We conducted a search in Pubmed, Embase, Web of Science, and Google Scholar databases from conception until October 2021 with keywords "microbotox" OR "mesobotox" OR "intradermal injection" AND "botulinum toxin" AND "rejuvenation" AND "wrinkle" AND "face-lift."

Results: Twenty articles were included in this review article. Different types of formulation including onabotulinum toxin A (OBA), abobotulinum toxin A (ABO), and incobotulinum toxin A (IBA) were used. The most used concentrations were 10–20 u/cc of OBA. Number of injection points varied from 4 (forehead and periorbital areas) to 999 (entire face) with 2 mm to 2 cm apart. Adjuvant treatment modalities were cross-linked hyaluronic acid, intense pulsed laser, and Fraxel laser. Most studies reported the initial results appeared after 5–14 days of procedure, lasting for nearly 3–6 months.

Conclusion: Microbotox is an efficient and attractive method for facial rejuvenation, mid-lower face-lifting, and fine wrinkles reduction in forehead, periocular, and cheek regions, especially in younger-aged subjects. Furthermore, it is a suitable treatment for neck rejuvenation and recontouring of lower mandibular border, particularly in older subjects with marked skin laxity.

KEYWORDS
botulinum toxin, face-lift, mesobotox, microbotox, rejuvenation

1 | INTRODUCTION

Botulinum toxin A (BoNTA) injection is the most popular non-invasive aesthetic procedure used to relax hyperkinetic musculature in face and neck. It blocks release of neurotransmitters such as acetylcholine (Ach), substance P, noradrenaline, vasoactive intestinal peptide (VIP), and calcitonin gene-related peptide (CGRP) at the neuromuscular junction.[1–26] Advantages are no downtime, rapid onset of effect, and high patient satisfaction. Microbotox or mesobotox is a relatively novel injection technique that characterizes as multiple intradermal microinjections of diluted BoNTA.[1–6] Injections accompany some resistance and lead to blanched wheal formation. This technique has more natural appearance compared to frozen facial expression revealed after conventional injection method of

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BoNTA.[1–6] In fact in this technique, each droplet of BoNTA is divided to several microdroplets, injected superficially which blocks only superficial expression muscles of face and neck. Furthermore, this injection method has more safety profile in danger zones or difficult-to-treat regions such as lower forehead, infraorbital, and mid and lower face with less complications such as eyebrow ptosis, blepharoptosis, ectropion, dry eye, asymmetric smile, mouth corners dropping, and swallowing impairment.[1–26] As no review article exists about the best injection techniques (dosage, concentration, as well as number, and location of injection points), efficacy and side effects of this innovative injection method of BoNTA, we decided to do so.

2 | METHODS

We conducted a search in Pubmed, Embase, Web of Science, and Google Scholar databases for keywords "microbotox" OR "mesobotox" OR "intradermal injection" AND "BoNTA" AND "rejuvenation" AND "wrinkle" AND "face-lift." We also sought bibliographies of the searched literature for relevant articles. Eligible criteria were those articles evaluating efficacy of microbotox technique in facial and neck rejuvenation or face-lift and wrinkle reduction. Only English articles including randomized clinical trials, comparative and non-comparative studies, case series, and case reports from conception until October 2021 were included. Basic science articles, letters to editor, review articles, and literature that did not evaluate efficacy of treatment via one of subjective or objective measurement methods were excluded. We identified 189 articles that after omitting duplicated and non-English ones, 149 articles were collected. Two first authors (B.I and M. KH.) evaluated articles firstly based on title and abstract for eligibility; then, forty-five articles were assessed comprehensively. Ultimately, twenty articles were included in this review article. Figure 1 demonstrates our search strategy.

3 | RESULTS

Twenty articles including eight randomized control trials (RCTs), two split-face comparative studies, one crossover study, six prospective case series, and three retrospective case series evaluated efficacy of microbotox technique on facial rejuvenation, facial lifting, and wrinkle reduction. One thousand and fifty hundred and forty-five subjects aged between 18 and 72 were included in the review. Different types of formulation including OBA, ABO, and IBA were used. All studies reconstituted the BoNTA with preservative-free 0.9% normal saline. Various concentrations of BoNTAand ABO (2–40 and 50–71.5 u/cc, respectively) were used. The most used concentrations were 10–20 u/cc of OBA. The

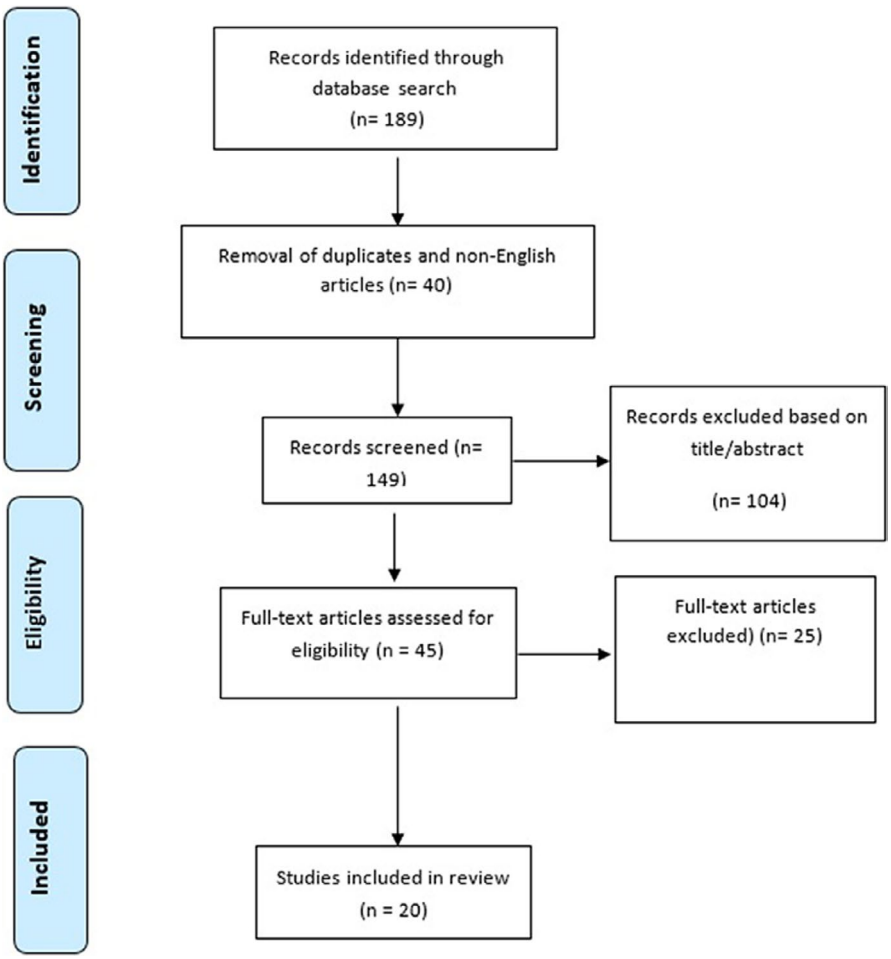


FIGURE 1 Schematic of the search strategy

majority of the studies used insulin syringe with 30–32 gauge needles. Furthermore, three studies used autoinjector and one study used 27 gauge microcannula for injection of areas with thin skin to avoid ecchymosis. In order to reduce discomfort, all of the studies used topical anesthesia; except one study that used nerve block. Furthermore, two studies mixed lidocaine to BoNTA formulation for more comfort. Number of injection points varied from 4 (forehead and periorbital areas) to 999 (entire face) with 2 mm to 2 cm interval. Injection volume in each point varied from 0.01 to 0.2 cc. Adjuvant treatment modalities included cross-linked hyaluronic acid (HA), intense pulsed laser (IPL), and non-ablative laser (Fraxel laser). Some studies reported the immediate results, while most of the studies reported the initial results appeared after 5–14 days of injection that lasted for 3–6 months. The results of literature are summarized in Table 1.

4 | EVALUATION METHODS

Both objective and subjective evaluation methods were used in the studies. Subjective methods included photonumeric scales, investigators global aesthetic improvement score (IGAIS), subjects global aesthetic improvement score (SGAIS), patient's satisfaction, visual analogue scale (VAS) for pain assessment and Merz's scale. Objective methods included brow height measurement, skin Visiometer, computerized image analysis, skin biopsy, immunohistochemistry (IHC), biophysical measurement (erythema index, melanin index, total epidermal water loss, and skin surfaces roughness), cutometer, sebumeter, and dermoscopic evaluation.

5 | DISCUSSION

Aging process is due to bone absorption, imbalance of elevator and depressor muscles, subcutaneous fat herniation, and skin laxity. Microbotox technique is used as an off-label method for facial lifting, skin tightening, softening of fine wrinkles, recontouring of cervicomental border, and improvement of soft-tissue ptosis, facial erythema, flushing, and skin texture due to reduction in sweat, skin oiliness, and pore size. The best candidates for this injection technique are younger patients with fine wrinkles looking for skin texture improvement and face-lift with more natural appearance. On the contrary, older subjects with loose skin and poor muscle tone are good candidates for cervicomandibular recontouring along with reducing of horizontal neck creases. Subjects with unrealistic expectations and hyperkinetic muscles are not suitable for this treatment.[1–26]

5.1 | Microbotox technique in upper-face region

Using microbotox technique in forehead area reduces rhytides while preventing eyebrow ptosis and unnatural appearance.[7–9] Three articles evaluated efficacy of ID injection of BoNTA (0.1–6 u/point in 4–60 injection points) for forehead wrinkling.[7–9] In two split-face RCTs, no significant differences were reported between immediate result of ID and IM injections of OBA (4–6 u/point, 2–5 injection points) in forehead area.[7,8] While one study revealed no difference between two techniques in efficacy longevity, the other one showed significantly less patients' satisfaction at 4-month follow-up with ID injection method.[7,8] In one retrospective case series, Zhang et al. found out significant improvement in forehead rhytides with combined microbotox and conventional techniques. In subjects with mild wrinkling, microbotox technique (OBA; 0.1 u/point, 40–60 injection points) was used for entire forehead. Whereas subjects with moderate-severe wrinkling were treated with conventional methods (total dose of 3–12 units at 6–10 points) in upper two-third areas and microbotox technique (0.2 u/point, 10–20 injection points) in lower-third areas. Touch-up sessions were performed only in 2.8% of cases of microbotox technique (total dose of 0.5–1.5 units).[9]

Glabellar wrinkles are usually more resistance to microbotox technique compared to forehead wrinkles.[15] Steinsapir et al. in one retrospective case series reported significant eyebrow elevation (average of 0.4 mm) with ID injection of OBA (0.33–0.66 u/point, 60–100 point) at the level of eyebrow surface and glabellar area. In order to avoid brow ptosis, injections should be limited to lower level of eyebrow cilia.[10]

Microbotox can rejuvenate periorbital region with less concern about complications including ectropion, dry eye, and exacerbation of eye bag or infraorbital wrinkling.[11,12] Two studies evaluated efficacy of microbotox in glabellar and periorbital regions. De Oliveira et al. evaluated efficacy of microbotox technique (OBA; 0.5 u/point, 21 points/side) in wrinkle reduction of infraorbital region. Higher percentage of improvement was revealed with subject evaluation compared to physician evaluation (86% and 66%, respectively). Thirty-four percent of cases required another sessions of injection with mean duration of 125 days.[11] Jeon et al. demonstrated significantly better efficacy of ID OBA injection (1 u/point, 5 injection points/side) in periocular rejuvenation compared to microneedling fractional radiofrequency (MFR) at 3- and 6-week follow-up while 3-month follow-up revealed superior efficacy of MRF compared to microbotox[12] (Figure 2).

5.2 | Microbotox technique in mid-lower face

Mid-lower face wrinkling, jowl formation, drooping of mouth corners, skin sagging, poor defined mandibular border, horizontal neck creases, and vertical bands of neck occur due to imbalance between elevator and depressor muscles. Weakening and relaxation of depressor muscles lead to face-lift, recontouring of the cervicomandibular angle, and restoration of youthful appearance due to action of unopposed elevator muscles. Furthermore, microbotox stimulates collagen production, reduces fibroblast degradation, improves skin elasticity, and also it decreases transepidermal water loss (TEWL) and skin surface roughness.[13–26]

TABLE 1 Results of microbotox injection method in literature

Reference	Study design	Number/ sex	Age (y)	Type of formulation	Dilution (u/cc)	Injection characteristics	Adjuvant therapy
de Quadros et al. (2018)[7]	Prospective RCT Split-face Single-blind	11F 5M	33 ± 5.96 (25–55)	OBA	NS	30 G needle Total volume:1 cc Total u/side: 26 Number of injection points/ side: 5	No
Gordin et al. (2014)[8]	Prospective RCT Split-face Double-blind	19 (NS)	NS	OBA	40	30 G needle Total u/side: 12 Number of injection points/ side: 2	No
Zhang et al. (2020)[9]	Retrospective Case series	341F 13M	41 ± 9 (27–61)	OBA	2–5	30 G needle Total volume:1–2 cc 0.03–0.05 cc/each point Number of injection points: 40–60 (entire forehead)/ 10–20 (lower forehead)	No
Steinsapir et al. (2015)[10]	Retrospective Case series	175F 52M	F:46 ± 4 M: 44 ± 8	OBA	33	32 G needle 0.01–0.02 cc/each point Total dose: 33 u Number of injection points: 60–100 Depth: 1 mm	No
de Oliveira et al. (2016)[11]	Prospective Case series	255F 45M	45 (18–72)	OBA	25	30G needle Total volume:0.88 cc Total dose in first/second sessions: 22/15 u Number of injection points: 6–24 (lateral)/6–12 (middle)/3–6 (medial) Distance: 0.5 cm	No
Jeon et al. (2013)[12]	Prospective Comparative Split-face Double-blind	12F	47 ± 4.44 (20–59)	OBA	10	30 G needle Total volume:0.5 cc Total dose: 5 u 0.1 cc/point Number of injection points/ side:5 Distance: 0.5 cm	No
Sapra et al. (2017)[13]	Prospective RCT Split-face Single-blind	10F	51.1 (44–60)	ABO OBA	OBA: 20 ABO: 50	Total volume/side:2.5 cc Total dose/side: 125 u (ABO), 50 u (OBA) Distance: 1 cm	No
Petchngaovilai et al. (2009)[14]	Retrospective Non-comparative	226F 49M	27–72	ABO	71.5	32 G needle Total volume:0.8–1 cc Total dose: 50–70 u/side Number of injection points: 34 Distance: 1 cm	No

Control group	Sites of injection	Indications	Evaluation methods	Results	S/E (prevalence/ duration)	F/U
Conventional technique	Forehead	Wrinkle reduction	Photograph Digital analysis Pt's satisfaction	No significant difference in efficacy and durability between two groups	No	12 w
Conventional technique	Forehead	Wrinkle reduction	Photograph Eyebrow height measurement Pt's satisfaction 5-point scale (pain)	No significant difference in wrinkle reduction, eyebrow elevation and Pt's satisfaction	Pain Tenderness (10.5%) Bruising (10.5%)	16 w
No	Forehead	Wrinkle reduction	Photograph Pt's satisfaction Brow position Wrinkle reduction (Merz forehead line scale) Pain (VAS)	No significant difference in brow position in four treatment groups Significant improvement in forehead wrinkles in four treatment groups compared to base-line	Slight stiffness (0.8%–42.5%, 1–2 w) Bruising (2.1%–5.4%)	2–4 w
No	Brow Glabella Crow's feet	Face-lift	Photograph Brow position Photonumeric scale	Significant improvement in eyebrow position, forehead wrinkles, and brow furrowing	Eyelid ptosis (0.2%, 2 w) Flu-like symptom (0.4%)	10–45 d
No	Infraorbital	Wrinkle reduction	Photograph Pt's satisfaction	Wrinkle improvement in 66% and 86% with Physician and subject evaluations, respectively	Pain (14%) Ecchymosis (4%)	5 mo
MFR	Periorbital	Wrinkle reduction	Photograph Pt's satisfaction IHC	Significant improvement in microbotox group at 3–6 w compared to MFR Significant improvement in MFR group at 18 w compared to microbotox group More increased in procollagen and elastin deposition in MFR compared to microbotox group	No	18 w
Conventional technique	Forehead Glabella Periorbital Cheek	Rejuvenation Texture improvement	Photograph Pt's satisfaction Visia Complexion Analysis System Image analysis of Sebutape	Significant improvement in mid–face-lift, forehead wrinkles and skin texture compared to base-line in microbotox groups No significant difference between two techniques and formulations	NS	16 w
No	Temporal Preuricular Jaw line Platysma	Face-lift	Photograph	High improvement (24.9%) Moderated improvement (65.52%) Minimal improvement (9.58%) in facial recontouring and wrinkle reduction	Minor asymmetry (3.07%)	14 w

(Continues)

TABLE 1 (Continued)

Reference	Study design	Number/ sex	Age (y)	Type of formulation	Dilution (u/cc)	Injection characteristics	Adjuvant therapy
Atwa et al. (2020)[15]	Prospective Placebo-control Split-face Double-blind	25F	39.8 ± 9.27 (30–50)	OBA	15	32 G needle Total volume/ side:0.3 cc 0.02 cc/point Number of injection points/ side: 15 Distance: 1 cm	No
Wanitphakdeedecha et al. (2016)[16]	Prospective RCT Split-face Double-blind	21F 1M	31.5 ± 5.5 (22–42)	ABO	71.5	31 G needle Total volume/side :1.6–2.5 cc Total dose: 120–180 u 0.04 cc/point Number of injection points/ side: 28	No
Seo et al. (2017)[17]	Prospective RCT Split-face Double-blind	24F	NS	OBA	10	Total volume/ side:1.5 cc Total dose: 15 u 0.05 cc/point Number of injection points/ side: 30	No
Zhu et al. (2017)[18]	Prospective RCT Double-blind	20F	40.5 (33–67)	OBA	12.5	Total volume:2.6 cc Total dose: 30 u 0.02 cc/point Number of injection points/ side: 120 Distance: 2 mm Depth:0.8–1 mm	No
Chang et al. (2008)[19]	Prospective RCT Split-face Double-blind	8F 1M	35–55	OBA	10	30 G needle Total volume/side: 2–2.5 cc Total dose: 20–25 u 0.02 cc/each point Number of injection points/ side: 100–125 Distance: 1 cm	No
Kapoor et al. (2010)[20]	Prospective RCT Placebo-control Split-face Double-blind	5F 5M	31.8 ± 3.1 (24–36)	OBA	20	30 G needle Total volume/ side:1.5 cc Total dose: 30 u 0.1 cc/point Number of injection points/ side: 30 Distance: 1 cm	No
Mole et al. (2014)[21]	Prospective Case series	40F 2M	54 (38–71)	OBA ABO	ABO: 62.5 ONA/ IBA: 25	27–30 G needle Total dose/ zone:8–16 u 0.02–0.04 cc/point Number of injection points: 15–20 Distance: 0.5 cm	Cross-linked HA
Kim et al. (2018)[22]	Prospective Case series	50F	49.8 (31–65)	IBA	13.3	32 G needle (auto injector) Total volume:1 cc 0.04 u/point Number of injection points: 990 Distance: 3.5 mm Depth:0.8–1 mm	Cross-linked HA

Control group	Sites of injection	Indications	Evaluation methods	Results	S/E (prevalence/ duration)	F/U
Saline	Forehead Glabella Crow's feet Nasolabial fold Marionette line Cheek Temporal Preuricular Jaw line	Face-lift	Photograph QGS Pt's self-assessment Pt's satisfaction	Significant improvement in forehead, nasolabial fold, and marionette lines with microbotox technique compared to placebo No significant difference between two groups in crow's feet improvement	Moderate discomfort (4%)	6 mo
Saline	Frontalis Corrugator supercilli Lateral orbicularis oculi Preuricular Jaw line Platysma	Face-lift	Photograph SGA PGA QGS	Significant better face-lift with microbotox technique compared to saline group	Mild brow ptosis (13.6%)	2 w
Saline	Cheek	Wrinkle reduction Texture improvement	Photograph Pt's satisfaction Visiometer clinician's evaluation	Significant improvement in wrinkles with microbotox technique in subject and clinician's evaluation No significant difference between two groups regarding other markers	NS	12 w
Saline	Entire face	Rejuvenation	Photograph Pt's satisfaction PGA Biophysical measurement (EI, MI, TEWL, elasticity, skin surfaces roughness)	Significant improvements in PGA, pt's satisfaction and biophysical markers (except EI and MI) in microbotox group compared to placebo	NS	12 w
Saline	Temporal Infraorbital Cheek Jaw line	Rejuvenation	Photograph Objective and subjective wrinkle assessments Pt's satisfaction Biopsy IHC	Significant improvement in wrinkle reduction in microbotox group compared to saline group No significant difference between two groups in face-lift and objective evaluation methods	Mild periorbital muscle weakness (11/.1%) Tolerable pain (100%)	16 w
Saline	Entire face	Rejuvenation	Photograph Sebumeter Pore size Skin-tightness Skin texture QGS	No significant difference between two sides of treatment	Tolerable pain	4 w
No	Frontal Temporal Periorbital Upper labial	Wrinkle reduction	Photograph	Excellent result in female subjects Failure: 7.1%	Transitory zygomatic palsy (2.4%,4 mo) Frozen look (19%)	1–25mo
No	Entire face	Rejuvenation Texture improvement	Photograph Dermoscope Biopsy TEWL Skin roughness and morphology	Significant improvement in fine wrinkle, TEWL and skin roughness	Ecchymosis (74%, 3–7 d) Lump(1–2 d)	6 mo

(Continues)

TABLE 1 (Continued)

Reference	Study design	Number/ sex	Age (y)	Type of formulation	Dilution (u/cc)	Injection characteristics	Adjuvant therapy
Kim et al. (2021)[23]	Prospective Case series	50F	46.7 (32–69)	OBA	13.3	31 G needle (auto injector) Total volume:1 cc 0.04 u/point Number of injection points: 999	Cross-linked HA
Bertossi et al. (2019)[24]	Prospective Non-comparative	38F 7M	43.8 (35–52)	OBA	20	30–32 G needle (autoinjector) Total volume:1 cc/side 0.01 u/point Number of injection points: 80–100	Cross-linked HA Fraxel laser
El Bedewi et al. (2012)[25]	Prospective Case series	10F	45 ± 10.3 (40–60)	OBA	10	30 G needle Injection angle: 45° Total volume:1.3 cc Total dose: 13 u 0.1 cc/each point Number of injection points:21 Distance: 2 cm	IPL
Awaida et al. (2018)[26]	Prospective Crossover Single-blind	25F	55.9 ± 5.8	ABO	70	30 G needle Total volume: 2–3 cc Mean dose: 154 ± 28.6 u Number of injection points: 150	No

Abbreviations: ABO, abobotulinum toxin A; d, day; EI, erythema index; F, female; F/U, follow-up; G, gauge; HA, hyaluronic acid; IBA, incobotulinum toxin A; IGAIS, investigator global aesthetic improvement scale; IHC, immunohistochemistry; IPL, intense pulse laser; M, male; MI, melanin index; mo, month; MRF, microneedling fractional radiofrequency; NS, not stated; OBA, onabotulinum toxin A; PGA, patients global assessment; Pt's, patients; QGS, Quartile grading scale; RCT, randomized controlled trial; S/E, side effect; SCH, stratum corneum hydration; SGA, subject global assessment; SGAIS, subjective global aesthetic improvement scale; TEWL, total epidermal water loss; u, unit; VAS, visual analogue scale; w, week; y, year.

Fourteen studies evaluated efficacy of microbotox technique (OBA; 0.04–1 u in 15–999 points) for facial rejuvenation and mid-lower face-lift.[13–26] Sapra et al. revealed more efficacy with microbotox technique compared to conventional method for mid–face-lifting. Furthermore, no significant difference was reported between efficacy of OBA and ABO in rejuvenation.[13] Petchngaovilai et al. in a retrospective study showed high-to-moderate improvement in mid–face-lifting and nasolabial softening in 90.4% of cases with microbotox technique (ABO; 2–4 u/point, 17 points/side).[14]

Six studies evaluated efficacy of microbotox technique compared to saline.[15–20] Four studies revealed significant improvement in face-lift and rhytides reduction with microbotox technique (OBA; 0.3 u/point, 15 points/side)/(ABO; 4–6 u/point, 28 points/side) compared to placebo.[15–18] One study showed higher percentage of improvement in older individuals (>35 years old) with poor muscle and lax skin.[15] In contrast, Wanitphakdeedecha et al. revealed significantly better results in younger subjects (<32 years) with chubby face appearance compared to older subjects with oval face appearance.[16] Two other RCTs showed no significant difference between intradermal injection of OBA (0.2–1 u/point, 30–125 points/side) and normal saline in mid-lower face regarding face-lifting and neocollagenesis using objective assessment methods.[19,20] However, subjective evaluation demonstrates good-to-excellent (more than 50%) improvement for wrinkling with microbotox in 44.4% of cases that initiates after 4 weeks of injection and persists for nearly 3 months.[19] They concluded that needling and mechanical stretch can stimulate neocollagenesis and skin tightening.[20]

Five studies evaluated impact of microbotox technique adjuvant with HA and lasers.[21–25] Three studies demonstrated significant impact of combined cross-linked HA with microbotox technique (ABO; 1–2 u/point, 15–20 points) or (OBA/IBA; 0.04–1 u/point, 15–999 points) for facial rejuvenation, wrinkle reduction, and skin hydration that lasted for 4–6 months.[21–23] Two other studies demonstrated efficacy of microbotox technique with ID injections of OBA (0.01–1 u/point, 21–100 points) combined with laser (IPL/Fraxel) with or without cross-linked HA in overall texture improvement, along with significant reduction in fine rhytides. Subjects with lighter skin types

Control group	Sites of injection	Indications	Evaluation methods	Results	S/E (prevalence/duration)	F/U
No	Entire face (except upper lid, eyebrows, lip, and hairline areas)	Wrinkle reduction Hydration	Photograph Dermoscope SGAIS Biopsy Topographic computer analysis TEWL SCH Skin roughness and morphology	Improvement in fine wrinkle, skin roughness (50.19%), and SCH (81.34%)	Lump(1 d) Ecchymosis (48%, 2–6 d)	6 mo
No	Glabellar Crow's feet Forehead Infraorbital	Rejuvenation	QGS Photograph	Significant improvement in skin tightness, oiliness, wrinkles, and brightness	No	12 mo
No	Forehead Cheek	Wrinkle reduction Texture improvement	Photograph Image analyzer computer system	Mild improvement in erythema, pore size, skin texture, telangiectasia, and fine wrinkles Significant improvement in frontoglabellar muscle contraction	No	1 w
Nefertiti technique	Lower face and neck area	Rejuvenation	Photograph Photonumeric scales IGAIS SGAIS Pt's satisfaction Pain (VAS)	Overall improvement: 84% via microbotox technique (versus 93.3% with Nefertiti technique) SGAIS: 100% with both techniques	Ecchymosis (12%) Less pain with microbotox compared to Nefertiti technique (VAS score 4.6 ± 2.3 vs. 0.6 ± 2.3)	2 w

(III and IV) and thinner skin showed higher improvement in erythema and fine wrinkles compared to those with darker skin phototypes and thicker skin.[24,25]

In one crossover study, Awaida et al. compared efficacy of microbotox technique (ABO; 0.9–1.4 u/point, 150 points) on lower face and neck rejuvenation compared to Nefertiti lift (deep IM injection into vertical neck bands and lower border of mandibular region). Microbotox technique demonstrated marked improvement in jowls and neck volume compared to Nefertiti technique, whereas Nefertiti technique showed higher improvement in vertical bands. Dysphasia was reported only with Nefertiti technique (4%) that resolved after 2 weeks[26] (Figure 2).

5.3 | Side effects of microbotox technique

Microbotox usually accompanies with higher pain perception during injections compared to conventional method. However, microbotox technique in forehead region generally has less pain compared to traditional method. Furthermore, microbotox technique usually leads to less bruising compared to conventional method due to superficial injection levels. Other side effects that have been reported in literature were slight stiffness feeling in forehead area for 1–2 weeks (0.8%–42.5%), eyelid ptosis for up to 2 weeks (0.2%), mild brow ptosis (13.6%), mild periorbital muscle weakness (11/.1%), flu-like symptom (0.4%), minor asymmetric smile for up to 4 months (3.07%), and frozen look (19%).[7–26]

6 | CONCLUSION

Good knowledge about musculature anatomy of face and neck, proper injection method (choosing suitable volume, concentration, dosage, and location of injection) and careful selection of patients are main factors to achieve optimized outcome with less adverse events with microbotox technique. Microbotox technique in forehead area usually leads to more natural appearance and less complications such as brow ptosis. Microbotox technique for entire

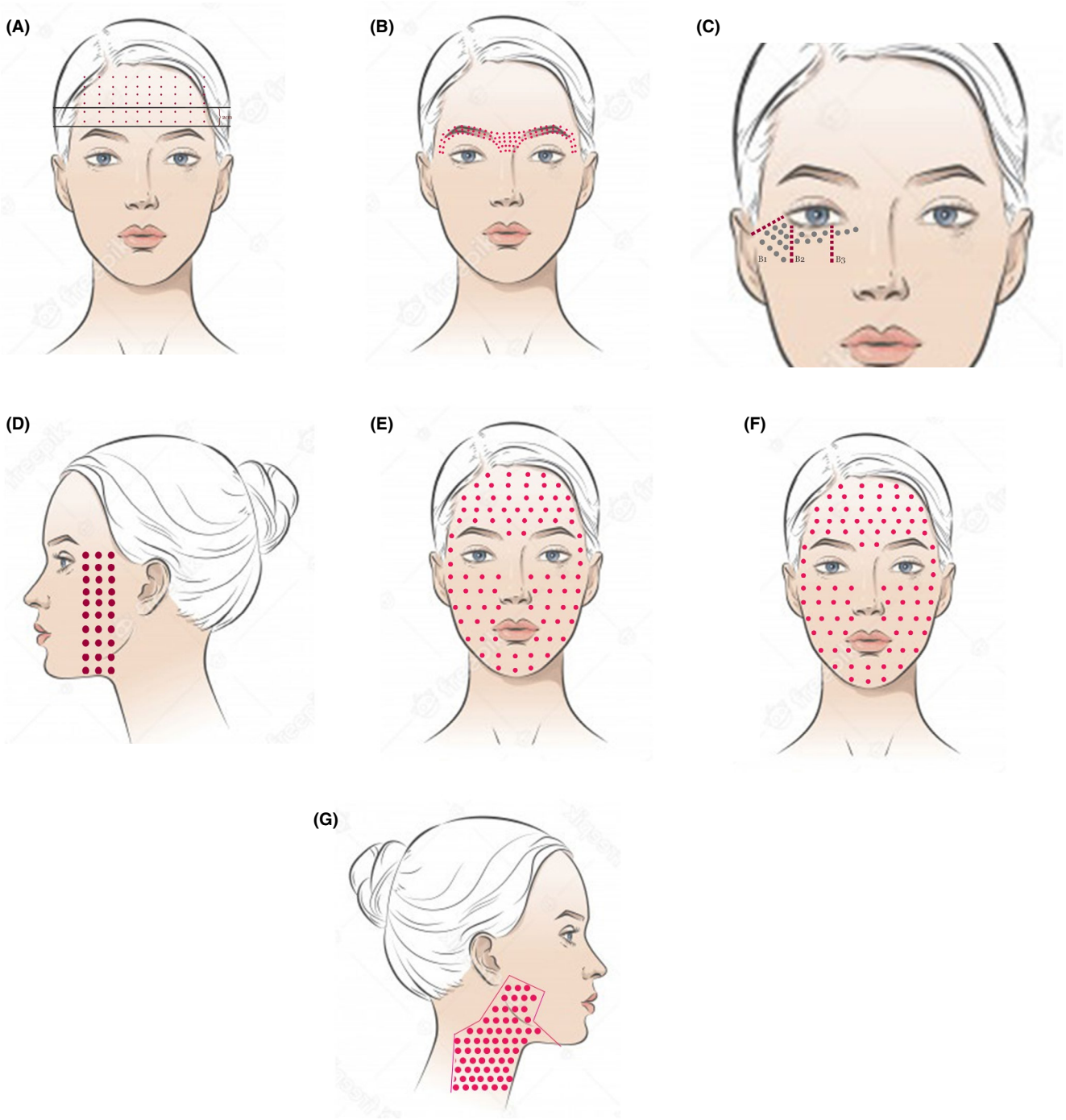


FIGURE 2 Illustrations demonstrating injected points with microbotox technique in included articles. (A) Entire forehead from lower hairline to upper eyebrow. (B) Eyebrows (lower border limited to the lowest brow cilia and lateral border limited to lateral raphe) as well as glabellar and upper nasal dorsum. (C) Lateral, middle, and medial parts of infraorbital region. (D) Forehead, glabellar, lateral parts of orbicularis oculi (one cm distance from orbital ridge) as well as platysma. (E) Temporal and cheek areas (from infraorbital to jaw line) (F) Entire face (G) Lower face and neck area (upper border: 5 cm above mandibular border, inner border: 1 cm distance from posterior of depressor anguli oris, outer border: anterior of sternocleidomastoid, inferior border: upper border of clavicle)

forehead area is a suitable method in subjects with fine wrinkling, while using conventional method in upper part of forehead along with microbotox technique in lower-third part might lead to better outcome in subjects with moderate-to-severe wrinkling. Employing microbotox technique in eyebrow and periocular areas can result in eyebrow elevation and periocular rejuvenation. In order to avoid brow ptosis, injections should be limited to lower level of eyebrow cilia. In glabella area, traditional method of injection has superior efficacy compared to microbotox technique. Furthermore, microbotox technique is a lunch-time procedure for mid-lower face-lift along with skin texture improvement in young individuals with fine wrinkling that want to avoid invasive face-lift methods. The best

candidate for recontouring of mandibular border and neck rejuvenation is older patients with skin laxity without hyperkinetic muscles. More RCTs are required to determine the best concentration, dosage and location of injection, and longevity of treatment.

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CONFLICT OF INTEREST

The authors of this manuscript do not declare conflicts of interest.

ETHICAL APPROVAL

The conducted literature review did not require the agreement of the bioethics committee.

AUTHORS CONTRIBUTION

B.I. and M. Kh contributed to the study conception and design. M.A. wrote the first draft of the manuscript. All authors performed material preparation, data collection, read and approved the final manuscript.

DATA AVAILABILITY STATEMENT

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

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